

ISAN 5367
Early Prediction of Sepsis in ICU Patients Using
Machine Learning
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Abstract

Sepsis is a potentially life-threatening condition triggered by the body's dysregulated response to infection, often resulting in tissue damage, organ failure, and death. Early diagnosis is critical but challenging due to nonspecific symptoms and rapid disease progression. This study explores a machine learning ML-based framework for early sepsis prediction using clinical time-series data. We apply various classification algorithms, including logistic regression, decision trees, and ensemble methods such as XGBoost (Chen), on a structured and preprocessed dataset. To handle the inherent class imbalance in sepsis data, we implement oversampling and undersampling techniques such as SMOTE (Nitesh). Model performance is evaluated using accuracy, recall, precision, and F1-score. Our unique contribution is the factorial comparison of model-balancing strategies within a reproducible, GPU-accelerated Google Colab environment. Results underscore the potential of data-driven decision support systems in enabling timely interventions and improving patient outcomes (Nemati) (Moor).

Summary Table

Aspect	Details
Project objective	Predict sepsis onset six hours before clinical diagnosis to enable earlier intervention and improve patient outcomes.
Database source	The PhysioNet/Computing in Cardiology Challenge 2019
Number of data points	1,552,210 hourly observations across all patients and time points
Number of patients	40336 unique patients
Number of variables	~40 clinical features including vitals, labs, and demographics
Prediction variable	Binary indicator of sepsis
How we cleaned it	Grouped by patient, applied forward/backward fill to propagate recent measurements, then median-imputed remaining gaps; removed septic-on-admission cases, dropped non-predictive columns, and encoded categoricals.
How many models	13 (e.g., logistic regression, KNN, decision tree, random forest, XGBoost, SVM, neural nets, and ensemble methods)
Hyperparameter tuning	Performed grid search with cross-validation (GridSearchCV) to identify optimal model parameters.
Best model	Undersampled XGBoost (Recallo = 0.86, Recall ₁ = 0.84; Accuracy = 0.86)

1. INTRODUCTION

1.1 Background & Motivation

Sepsis remains one of the leading causes of mortality in hospitalized patients, accounting for significant health burdens globally (WHO, 2020). Its early detection is crucial yet challenging due to its nonspecific symptoms. Predictive analytics and machine learning (ML) offer powerful tools for analyzing electronic health records (EHR) to flag early signs of sepsis (Shickel). Prior research indicates the successful use of ML in early disease detection, but gaps remain in the comparative analysis of sampling strategies and classifier performance in sepsis-specific datasets. The objective of this study is to evaluate the effectiveness of various ML models paired with data-balancing techniques in accurately predicting sepsis 6 hours before it happens. Our goal is to develop a model that is not only accurate but also interpretable and practical for real-world clinical applications.

Healthcare systems often struggle to identify sepsis in its earliest stages due to delays in lab test results and subjective assessments. The integration of ML algorithms with real-time monitoring tools provides an opportunity to automate risk scoring and generate alerts, potentially reducing time to treatment and improving outcomes. Moreover, as healthcare data grows in volume and complexity, ML models become indispensable tools in extracting clinically actionable insights (Shickel) (Choi).

Previous studies have demonstrated the use of ML to outperform traditional sepsis scoring systems such as SIRS and SOFA (Nemati). However, issues such as imbalanced datasets and the lack of interpretability remain. Our study aims to address these by combining various classification algorithms with data-balancing methods to build a predictive model that is robust, interpretable, and suitable for real-world deployment. All model development is conducted in Google Collab, ensuring scalability and reproducibility.

1.2 Literature Review

Several machine learning approaches have been explored for early sepsis prediction. Notably, (Nemati) developed a highly interpretable ML model using ICU data that showed high sensitivity. (Moor) reviewed the landscape of ML-based sepsis prediction, emphasizing the effectiveness of tree-based models and deep learning techniques. (Choi) introduced the RETAIN model that leverages reverse time attention for interpretability in healthcare settings.

(Shickel) conducted a comprehensive survey on deep learning in EHR analysis and identified sepsis as a promising area where such methods can yield transformative clinical insights. Their work highlights how RNNs and attention mechanisms can uncover temporal dependencies in patient records, which are crucial for early identification of sepsis progression.

Additionally, the use of ensemble learning methods, particularly Random Forests and Gradient Boosting Machines, has proven effective in managing high-dimensional clinical data. These models often outperform single learners by reducing variance and improving generalization (Moor). However, the tradeoff between performance and interpretability remains a recurring theme, especially in critical care contexts where decisions need to be transparent to clinicians.

Imbalanced data continues to be a challenge in this domain. As sepsis occurs in a minority of ICU cases, traditional models may bias predictions toward the majority class. This has led to the adoption of resampling techniques such as SMOTE (Nitesh) and adaptive synthetic sampling. Despite their popularity, these methods can sometimes introduce noise or lead to overfitting. Therefore, evaluating the impact of such techniques across a variety of ML models is essential for robust sepsis prediction (Fernández).

Feature selection and engineering are also pivotal in refining model accuracy and interpretability. For instance, time-dependent statistical summaries (e.g., moving averages, slopes) derived from vital signs can enhance predictive performance. Tools like SHAP (Lundberg) further facilitate transparency by attributing predictions to individual features, enabling medical professionals to understand the rationale behind alerts.

SMOTE remains a standard for handling class imbalance (Nitesh), while tree-based models such as XGBoost continue to show promising results in structured data settings (Chen). Our contribution lies in systematically comparing these combinations using stratified cross-validation, feature engineering, and interpretability tools like SHAP to propose an optimized framework for real-world application in ICUs.

1.3 Novelty & Objectives

The novelty lies in:

The novel aspects of our study include:

- Conducting all ML workflows in Google Collab for efficient, GPU-powered iteration.
- Implementing 13 different algorithms paired with both oversampling and undersampling techniques to build a comprehensive model-balancing matrix.
- Ensuring data integrity through Patient_ID-based cross-validation to avoid data leakage.
- Visualizing model performance tradeoffs and feature contributions using SHAP (Lundberg).

Objectives

- To preprocess and structure time-series EHR data.
- To address class imbalance through advanced resampling techniques.
- To evaluate and fine-tune a diverse set of classification algorithms.
- To interpret model outputs in a clinically actionable manner.
- To support clinicians with timely risk predictions and model explainability.

1.4 Scope & Report Structure

The remainder of this report is organized as follows: Section 2 outlines the research design and data preparation; Section 3 covers data analysis and model training; Section 4 presents experimental results; Section 5 discusses the findings and their clinical implications; Section 6 concludes with future work. A comprehensive reference list is included at the end of the document.

2. RESEARCH DESIGN

2.1 Dataset Description

The dataset used in this project is derived from ICU patients' electronic health records (EHR), which contain time-series clinical parameters such as vital signs and laboratory test results. Each observation is assigned a binary outcome (SepsisLabel: 0 for non-sepsis, 1 for sepsis), resulting in a supervised classification issue. The data was loaded and analyzed in Google Colab using Python-based frameworks, which enabled repeatability and scalability testing.

The features fall into 4 categories

1. Vitals (Real-Time Monitoring Variables)-

- HR | Heart rate | beats per minute
- O2Sat | Oxygen saturation via pulse oximetry | %
- Temp | Body temperature | degrees Celsius
- SBP | Systolic blood pressure | mmHg
- MAP | Mean arterial pressure | mmHg
- DBP | Diastolic blood pressure | mmHg
- Resp | Respiratory rate | breaths per minute
- EtCO2 | End-tidal CO2 | mmHg (optional)

2. Laboratory Measurements

- BaseExcess | Excess base in blood | mmol/L
- HCO3 | Bicarbonate level | mmol/L
- FiO2 | Fraction of inspired oxygen | %
- pH | Blood pH level
- PaCO2 | Arterial CO2 pressure | mmHg
- SaO2 | Arterial oxygen saturation | %
- AST | Aspartate aminotransferase (liver enzyme) | IU/L
- BUN | Blood urea nitrogen | mg/dL
- Alkalinephos | Alkaline phosphatase | IU/L
- Calcium | Calcium level | mg/dL
- Chloride | Chloride level | mmol/L
- Creatinine | Kidney function marker | mg/dL
- Bilirubin_direct | Direct bilirubin | mg/dL
- Glucose | Blood glucose | mg/dL
- Lactate | Lactic acid concentration | mmol/L

- Magnesium | Magnesium level | mmol/L
- Phosphate | Phosphate level | mg/dL
- Potassium | Potassium level | mmol/L
- Bilirubin_total | Total bilirubin | mg/dL
- TroponinI | Cardiac marker | ng/mL
- Hct | Hematocrit | %
- Hgb | Hemoglobin | g/dL
- PTT | Partial thromboplastin time | seconds
- WBC | White blood cell count | count/L
- Fibrinogen | Fibrinogen level | mg/dL
- Platelets | Platelet count | count/mL

3. Demographics / Administrative

- Age | Age of the patient (in years)
- Gender | 0 = Female, 1 = Male
- Unit1 | 1 if in MICU (Medical ICU), else 0
- Unit2 | 1 if in SICU (Surgical ICU), else 0
- HospAdmTime | Time between hospital & ICU admission (hours)
- ICULOS | ICU length of stay at current time step

4. Target Variable

- SepsisLabel 1 if the patient is in a window 6 hours before sepsis onset, else 0.

Figure 1: Snapshot of the original Dataset

2.2 Descriptive Statistics

Before modeling, we first summarized the central tendency and dispersion of all candidate predictors. Our full cohort consists of 1,552,210 hourly observations drawn from 40336 patients; after grouping and filtering, the working dataset retained the same number of rows but varying counts per variable depending on missingness.

Attribute	count	mean	std	min	25%	50%	75%	max
Hour	1552210	25.492745	28.88256	0	9	19	33	335
HR	1398811	84.581443	17.32524	20	72	83.5	95.5	280
O2Sat	1349474	97.193955	2.936924	20	96	98	99.5	100
Temp	525226	36.977228	0.770014	20.9	36.5	37	37.5	50
SBP	1325945	123.750465	23.23156	20	107	121	138	300
MAP	1358940	82.4001	16.34175	20	71	80	92	300
DBP	1065656	63.830556	13.95601	20	54	62	72	300

Resp	1313875	18.726498	5.098194	1	15	18	21.5	100
EtCO2	57636	32.957657	7.951662	10	28	33	38	100
BaseExcess	84145	-0.689919	4.294297	-32	-3	0	1	100
HCO3	65028	24.075481	4.376504	0	22	24	26.8	55
FiO2	129365	0.554839	11.12321	-50	0.4	0.5	0.6	4000
pH	107573	7.378934	0.074568	6.62	7.34	7.38	7.43	7.93
PaCO2	86301	41.021869	9.267242	10	35	40	45	100
SaO2	53561	92.654188	10.89299	23	94	97	98	100
AST	25183	260.223385	855.7468	3	22	41	111	9961
BUN	106568	23.915452	19.99432	1	12	17	28	268
Alkalinephos	24941	102.483661	120.1227	7	54	74	108	3833
Calcium	91331	7.557531	2.433152	1	7.7	8.3	8.7	27.9
Chloride	70466	105.82791	5.880462	26	102	106	109	145
Creatinine	94616	1.510699	1.805603	0.1	0.7	0.94	1.43	46.6
Bilirubin_direct	2990	1.836177	3.694082	0.01	0.2	0.445	1.7	37.5
Glucose	265516	136.932283	51.31073	10	106	127	153	988
Lactate	41446	2.646666	2.526214	0.2	1.26	1.8	3	31
Magnesium	97951	2.05145	0.397898	0.2	1.8	2	2.2	9.8
Phosphate	62301	3.544238	1.423286	0.2	2.6	3.3	4.1	18.8
Potassium	144525	4.135528	0.64215	1	3.7	4.1	4.4	27.5
Bilirubin_total	23141	2.114059	4.311468	0.1	0.5	0.9	1.7	49.6
TroponinI	14781	8.290099	24.80624	0.01	0.04	0.3	3.98	440
Hct	137433	30.794093	5.491749	5.5	27	30.3	34.1	71.7
Hgb	114591	10.430833	1.968661	2.2	9.1	10.3	11.7	32
PTT	45699	41.231193	26.21767	12.5	27.8	32.4	42.8	250
WBC	99447	11.446405	7.731013	0.1	7.6	10.3	13.8	440
Fibrinogen	10242	287.385706	153.0029	34	184	250	349	1760
Platelets	92209	196.013911	103.6354	1	126	181	244	2322
Age	1552210	62.009469	16.38622	14	51.68	64	74	100
Gender	1552210	0.559269	0.496475	0	0	1	1	1
Unit1	940250	0.496571	0.499989	0	0	0	1	1
Unit2	940250	0.503429	0.499989	0	0	1	1	1
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HospAdmTime	1552202	-56.125122	162.2569	5366.86	-47.05	-6.03	-0.04	23.99
ICULOS	1552210	26.994992	29.00542	1	11	21	34	336
SepsisLabel	1552210	0.017985	0.132896	0	0	0	0	1
Patient_ID	1552210	59201.47971	50248.19	1	9990	19965	109878	120000

Table 1: Descriptive Statistics of the Dataset

1. Vital signs.

- Heart rate (HR) averages 84.6 bpm (SD 17.3; min 20, max 280) over ~1.4 million records.
- Oxygen saturation (O₂Sat) is tightly distributed (mean 97.2 %, SD 2.94; min 20, max 100) across ~1.35 million readings.

- Blood pressure measures (SBP, MAP, DBP) show a means of 123.8 mmHg (SD 23.2), 82.4 mmHg (SD 16.3), and 63.8 mmHg (SD 14.0) respectively, each with over one million entries but extreme maxima of 300 mmHg.
- Respiratory rate (Resp) has mean 18.7 breaths/min (SD 5.1), min 1, max 100, suggesting occasional recording errors or extreme physiology.
- Temperature (Temp) appears normally centered on 36.98°C (SD 0.77) from ~525 K measurements, with a few implausible high values up to 50 °C.

2. Laboratory tests.

- Standard chemistries (creatinine, chloride, calcium, magnesium, phosphate, potassium) and blood counts (WBC, Hct, Hgb, platelets) exhibit plausible clinical ranges and low-to-moderate variability (e.g., creatinine mean 1.51 mg/dL, SD 1.81; platelets mean $196 \times 10^3/\mu\text{L}$, SD 104).
- Some analytes show heavy right-tails or outliers: AST max 9,961 U/L (mean 260, SD 856), FiO₂ max 4,000 % (mean 0.55, SD 11.1), and lactate max 31 mmol/L (mean 2.65, SD 2.53).

3. Demographics & metadata.

- Age is well-recorded for all 1.55 M observations (mean 62.0 years, SD 16.4; range 14–100).
- Gender is binary coded (mean 0.56, SD 0.50).
- Unit identifiers (Unit1, Unit2) each appear in ~940 K rows (mean ≈ 0.50), confirming roughly equal distribution between the two ICUs.
- Admission timing (HospAdmTime) averages –56 hours (SD 162) relative to ICU entry, with extreme negative values (min –5,366 h) reflecting data-entry or transfer timing anomalies.
- ICU length of stay (ICULOS) averages 27 days (SD 29; min 1, max 336).

4. Target variable.

- Sepsis labels are rare: mean 0.018 (1.8 % positive rate), underscoring the need for imbalance-aware modeling.

These summaries guided our preprocessing choices: first we use forward & backward filling to fill some of the missing data for the same patient id, we capped implausible outliers in vitals and labs, used median imputation for numeric features (to mitigate skew and extreme values). After imputation and outcome-window extraction, we removed several metadata fields that do not contribute to the 6-hour sepsis prediction task Unit1 & Unit2- These binary indicators specify which ICU ward the patient was admitted to. Because our modelling goal is to generalize across care units—and since unit assignment may reflect hospital logistics rather than patient physiology—we exclude them to avoid spurious correlation. We have also dropped ICULOS later in the code right before implementing the machine learning models.

2.3 Missing Data Handling

A thorough review of feature completeness reveals three distinct patterns of missingness (Figure 2). First, the time-varying vital signs—Hour (≈ 10 % complete), heart rate (HR, ≈ 13 %), oxygen saturation (O₂Sat, ≈ 13 %), systolic blood pressure (SBP, ≈ 14 %), diastolic blood pressure (DBP, ≈ 12 %), mean arterial pressure (MAP, ≈ 12 %), respiratory rate (Resp, ≈ 15 %) and temperature (Temp, ≈ 66 %)—are heavily under-reported, reflecting irregular charting or recording gaps. Second, many laboratory measurements (e.g., EtCO₂, BaseExcess, HCO₃, PaCO₂, pH, AST, BUN, alkaline phosphatase, calcium, chloride, creatinine, bilirubin direct/total, lactate, magnesium, phosphate, potassium, troponin I, hematocrit, hemoglobin, PTT, WBC, fibrinogen, platelets) exhibit very high completeness (≈ 93 –100 %), indicating reliable lab-test availability. One notable exception is serum glucose (≈ 83 %), which may require careful handling due to its clinical importance. Third, static demographics and metadata—age, gender, care unit indicators (Unit1/Unit2, each ≈ 40 %), hospital admission time (HospAdmTime), ICU type, and identifier columns (Patient_ID, SepsisLabel)—show either partial or near-zero completeness by design (identifiers intentionally removed from modeling features).

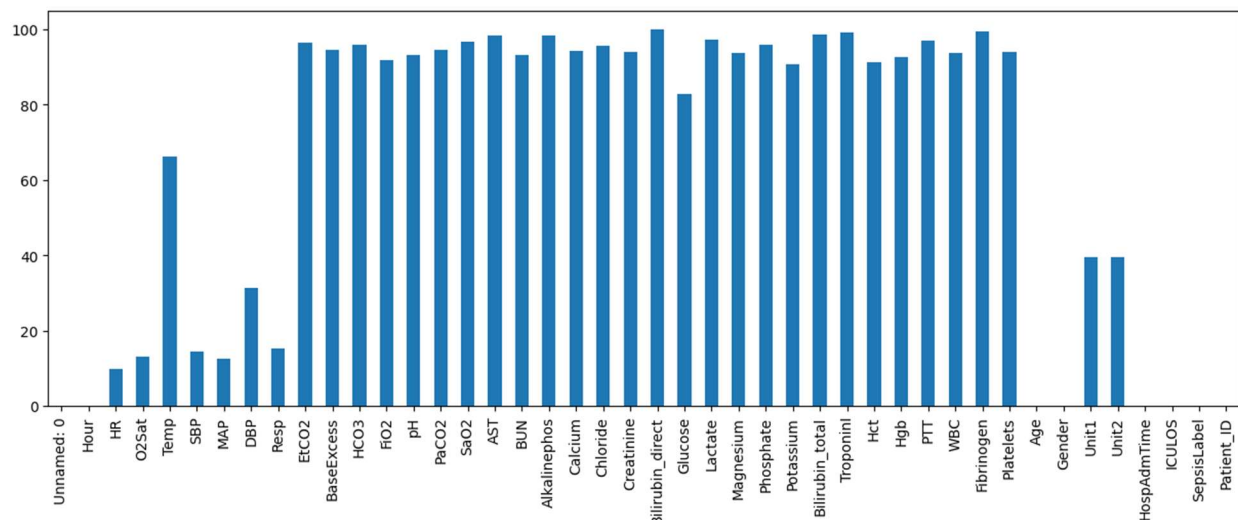


Figure 2: Missing Data Bar Chart

To address these gaps, we imputed numeric labs and vital signs using median values (after confirming non-skewed distributions), filled categorical indicators (Unit1/Unit2) with zeros to denote absence, and excluded features with insufficient coverage for reliable modeling (e.g., Hour). This strategy preserves maximal information from well-reported variables while mitigating bias from sporadic observations.

3. Data Analysis & Results

3.1 Data Preparation

3.1.1 Target Variable Distribution

Before modeling, we examined the distribution of our binary outcome (SepsisLabel) and its subtypes (sepsisType) to understand class imbalance and the timing of sepsis events.

- *SepsisLabel(ovs.1)*

As shown in Figure 3, the vast majority of hourly observations are labeled “no sepsis” (0), with only about 1.8 % of records flagged as sepsis (1). Concretely, ~1.52 million entries have SepsisLabel = 0, versus only ~30 thousand with SepsisLabel = 1. This extreme imbalance underscores the need for specialized sampling methods during training to avoid a trivial “all-negative” classifier.

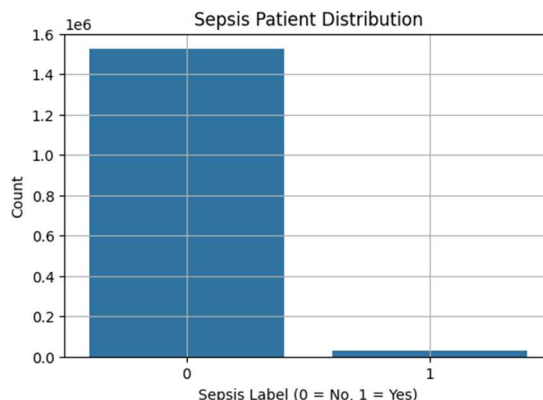


Figure 3: Sepsis Patient Distribution

- *Sepsis Type breakdown*

When we further categorize the positive cases by sepsis type (Figure 4), we see that:

- *NonSepsis* (no sepsis) again dominates with ~1.38 million records.
- *SepsisAfterAdm* comprises the bulk of sepsis events (~170 thousand records), reflecting cases where sepsis onset occurs during the ICU stay.
- *SepsisBeforeAdm* is extremely rare (<5 thousand records), indicating few patients admitted already septic.

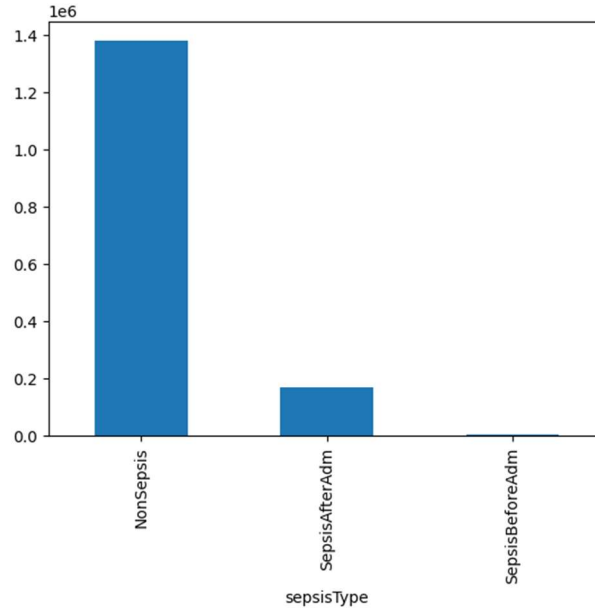


Figure 4: *SepsisType Distribution*

In response, our pipeline applies SMOTE oversampling on the minority class and random under-sampling of the majority class—stratified by patient—to ensure the model sees enough positive examples without overfitting to noise.

3.1.2 Imputation & Outcome-Window Extraction

To prepare each patient’s data for “6-hour-ahead” sepsis prediction, we first handled missing values and then distilled a single representative row per patient:

1. *Median Imputation of Remaining Gaps:* For any feature still containing nulls after the initial fill steps, we computed its overall median and replaced missing entries in place. Using the median guards against extreme values and preserves the central tendency of each variable.
2. *Excluding Already-Septic Admissions:* We dropped all patients whose very first ICU record (Hour = 0) was already labeled septic. Since our goal is “early” detection within the ICU, removing these cases avoids conflating prediction with post-admission diagnoses.
3. *Extracting the 6-Hour-Prior Snapshot:*
 - *Septic Patients:* For each patient who eventually develops sepsis, we locate the earliest sepsis onset time (tsepsis), subtract six hours, and select that row’s measurements as our model input. If an exact “-6 h” timestamp is missing, we fallback to the last available record before onset.
 - *Non-Septic Patients:* We uniformly select the row at 6 h before the end of their stay (or the final record if their stay is shorter), and label it as non-septic.

This yields one row per patient—each tagged with SepsisLabel indicating whether sepsis occurs within the next six hours—ideal for training and evaluation.

Unnamed: Hour	HR	O2Sat	Temp	SBP	MAP	DBP	Resp	EtCO2	BaseExcess	HCO3	PiO2	pH	PtcO2	SpO2	AST	BUN	Alkalineph	Calcium	Chloride	Creatinine	Bilirubin	Glucose	Lactate	Magnesium	Phosphate	Potassium	Bilirubin	Troponin	Hct	Hgb	PTT	WBC	Fibrinogen	
48	48	100	85.5	36.33	161	104	36	33	19	48	0.25	7.32	100	89	16	22	98	9.6	85	0.7	0.3	133	1.52	2.2	3.7	4.6	0.3	0.11	36.2	12.2	30.7	14.7	275	
17	17	95	97	36	114	92	36	33	0	22	0.45	7.4	40	97	33	100	71	7.8	113	2.5	0.3	78	1.52	2.5	4.4	5.1	0.8	0.11	27.8	9.7	30.7	11	275	
42	42	72	97	37.06	151	92	63	28	33	5	29	0.8	7.49	38	97	33	25	71	11	100	0.8	0.3	130	1.52	2.4	2.9	4.1	0.8	0.11	32.1	11	29.5	8.7	275
23	23	107	97	36.39	125	62.33	51.5	15	33	0	22	0.45	7.41	41	98	33	19	71	8.2	108	0.8	0.3	69	1.52	2.4	3.8	4.3	0.8	0.11	24	8.3	22.3	7.6	275
42	42	80	98	37.28	141	93	62	14	33	0	28	0.45	7.4	40	97	16	5	62	8.2	106	0.7	0.3	103	1.52	2.5	2.8	3.6	0.6	0.11	39.7	14.4	29	9	275
11	11	87	99	36.72	122	80	62	43	33	0	29	0.4	7.34	47	97	33	9	71	8.4	111	0.7	0.3	73	1.4	2	3.4	3.8	0.8	0.11	36.9	12.2	30.7	12	275
39	39	112	96	38.39	111	76	61.5	14	33	-9	13	0.4	7.33	27	97	462	64	88	7.3	115	3.9	0.3	262	2.2	1.6	3.5	3.9	1.4	0.11	40.2	14.5	27.1	8.8	275
34	34	68	100	36	108	62	44	17	33	-8	15	0.45	7.36	26	97	33	31	71	7.9	105	1.2	0.3	108	1.5	1.9	3.5	5	0.8	0.11	25	8.5	30.7	8.4	275
242	242	135	97	38.67	136	101	82	31	33	1	23	0.35	7.45	37	97	33	22	71	7.5	112	0.7	0.3	128	1.5	2.4	2	4	0.8	0.11	28.5	9.2	24.4	13.6	477.5
17	17	83	96	37.3	111	72	55	17	33	0	25	0.4	7.4	38	96	33	17	71	8.4	105	1.1	0.3	92	1.1	2.1	3.4	3.7	0.8	0.11	32.8	10.9	29.9	9.9	275
18	18	94	100	37.63	142	87	61	18	33	0	23	0.4	7.42	36	97	33	17	71	8.5	109	1	0.3	147	1.1	2.4	3.9	3.8	0.8	0.11	33.2	12.2	24.9	10.6	275
15	15	100	95	37.28	121	20	62	16	33	0	22	0.45	7.4	40	97	33	32	71	7.8	115	0.9	0.3	88	1.52	1.9	2.9	3.9	0.8	0.11	26.5	9	25.8	7.3	275
33	33	90	97	36.72	112	86.67	62	22	33	0	22	0.45	7.38	40	97	8	19	70	7.3	103	0.8	0.3	65	1.5	1.4	2.4	4.1	0.5	0.11	31	10.5	53.9	12.7	275
36	36	100	99	37.3	106	72	49	15	33	0	27	0.5	7.39	42	96	33	16	71	8.4	108	1.1	0.3	115	2.3	2	2.8	3.6	0.8	0.11	32.5	11.4	34	14.7	275
14	14	87.5	98	36.7	103.5	80.5	66	13	33	-9	17	0.4	7.31	31	99	33	29	71	7.7	105	4.4	0.3	95	1.6	1.8	5.3	4.5	0.8	0.11	29.5	9.9	35.6	17.7	275
13	13	106	95.5	37.8	105	59.5	104	17.5	33	-5	23	0.5	7.35	34	70	33	13	71	8.4	113	1.3	0.3	87	1.52	2.3	3.4	4.7	0.8	0.11	31.1	10.8	28.1	12	275
31	31	70	100	36.72	131	81.67	62	18	33	0	24	0.45	7.4	40	97	33	11	71	8.9	108	0.7	0.3	119	3.6	2.2	2.5	4	0.8	0.11	34.6	11.7	30.7	6.2	275
119	119	100	96	37.61	123	87	80	22.5	33	5	28	0.4	7.43	46	97	54	9	97	8.1	102	0.3	0.3	147	1.1	2	3.5	4	1.1	0.11	24	7.9	30.7	12.3	275
42	42	61	100	36.4	163	95	69	16	33	-10	19	0.4	7.24	38	97	33	87	71	8.4	96	8.7	0.3	88	0.9	1.9	3.4	5.7	0.8	0.3	30.6	10.7	36.8	2.5	275
13	13	65	97	36.89	124	78	62	12	33	0	30	0.45	7.4	40	97	33	11	71	8.3	102	1	0.3	108	1.52	2.3	4.3	4.5	0.8	0.11	35.8	10.5	26.4	8.3	275
51	51	109	92	36.67	123	83	57	29	33	0	25	0.45	7.42	37	97	12	8	55	7.4	103	0.4	0.3	83	1.52	1.7	2.1	3.9	0.6	0.11	27.9	9.2	36.2	16.4	275
3	3	80	99	36.8	89	57	41	18	33	-3	21	0.5	7.37	36	88	33	34	71	8.4	112	0.7	0.3	129	1.52	2.4	3.4	4.3	0.8	0.11	32.8	11.4	30.7	25.1	275
15	15	84	100	37.5	139	97	62	18	33	0	24	0.45	7.4	40	97	33	65	71	8.9	96	7	0.3	238	1.52	2.5	10.5	6.1	0.8	0.11	34.7	11.1	30.3	12.5	275
45	45	84	100	36.22	157	92.33	62	20	33	0	19	0.45	7.4	40	97	33	33	71	7.1	111	1.1	0.3	179	1.52	1.9	3.1	4.1	0.8	0.11	33.1	11.4	42.8	11.4	275
44	44	69	97	36.78	90	63.33	62	16	33	0	24	0.45	7.4	40	97	33	10	71	8.4	104	0.9	0.3	92	1.52	2	3.4	4	0.8	0.11	39.6	13.4	63.8	10.3	275
7	7	108	97	36.11	88.5	56	69	23	33	0	25	0.45	7.4	40	97	33	19	71	8.3	109	0.9	0.3	140	1.52	1.7	4.2	4.3	0.8	0.11	29.1	10.1	36.3	17	275
4	4	71	96	36.89	117	71.67	62	19	33	0	24	0.45	7.4	40	97	33	17	71	8.4	106	0.9	0.3	122	1.52	2	3.4	4	0.8	0.11	31.4	10.5	30.7	10.2	275
15	15	84	100	37.06	122	101	88	32	33	-4	20	0.55	7.38	33	97	28	27	283	7.5	113	0.6	0.3	279	1.5	2.3	3.2	4.4	1.1	0.11	30.8	9.9	26.1	10.9	549
42	42	68	100	36.72	131	90	76	23	33	0	23	0.45	7.4	40	97	33	9	71	8.8	106	0.6	0.3	109	1.52	2	2	3.5	0.8	0.11	28.8	9.5	25.8	15.2	275
39	39	68	98	36.44	94	69	55	19	33	2	25	0.5	7.45	38	95	33	16	71	8.3	102	1	0.3	115	0.8	2.2	3.7	4	0.8	0.11	22.3	7.5	31.3	6.2	275
49	49	74	98	37	133	67	50	16	33	0	24	0.45	7.4	40	97	15	9	126	7.6	104	0.6	0.3	126	1.52	2	1.9	3.8	0.5	0.11	37.6	13.5	25.4	10.1	275
45	45	62	97	36.56	106	92	62	25	33	0	25	0.45	7.4	40	97	33	25	71	8.2	113	0.8	0.3	109	1.52	1.8	2.2	3.9	0.8	0.11	30.6	10.4	25.7	11.6	275

Figure 5: Dataset After Cleaning

3.2 Model Implementation

Following comprehensive preprocessing, the dataset was divided into training and validation sets using an 80:20 stratified split based on the SepsisLabel to ensure proportional representation of septic and non-septic cases in both sets. Due to the pronounced class imbalance—approximately 7% of patients labeled as septic—resampling techniques were applied to the training data. These included oversampling using the Synthetic Minority Oversampling Technique (SMOTE) (Nitesh), which synthetically generates new instances of the minority class, and undersampling via random selection of the majority class to balance the dataset.

Normalization techniques were applied based on model requirements. Distance-based models such as K-Nearest Neighbors (KNN) and neural networks were normalized using MinMax scaling to rescale all features between 0 and 1, whereas linear models like Logistic Regression and Support Vector Machines (SVM) were standardized using the z-score method via StandardScaler (fabian2011scikit).

We implemented thirteen classification models, encompassing both traditional and ensemble learning approaches. Baseline classifiers included Logistic Regression, Decision Tree, Gaussian Naive Bayes, and SVM. Advanced classifiers comprised ensemble methods such as Bagging, AdaBoost, Gradient Boosting, Random Forest, and XGBoost (Chen), in addition to neural networks configured as shallow feed-forward multilayer perceptrons. KNN was also evaluated due to its interpretability and sensitivity to feature scaling.

Hyperparameter tuning was conducted using GridSearchCV with 5-fold cross-validation for key models. Logistic Regression was tuned over a grid of regularization strengths ($C \in [0.01, 0.1, 1, 10, 100]$), solvers (lbfgs, saga, newton-cg), and penalty='l2'. Decision Trees were optimized over parameters such as max_depth, min_samples_split, and min_samples_leaf. Ensemble models were tuned for number of estimators, maximum depth, and learning rates (where applicable), though initial XGBoost training was performed using default parameters.

Cross-validation folds were stratified and grouped by Patient_ID to prevent data leakage from temporally correlated observations within the same patient. Model performance was evaluated using metrics crucial to clinical decision-making: class-wise recall, precision, F1-score, and overall accuracy. This methodology ensured robust model comparisons under varying imbalance conditions, enabling us to identify the best-performing classifier in both sensitivity and specificity for early sepsis prediction.

3.3 Performance Results

This section presents a comparative performance analysis of thirteen machine learning classifiers under three sampling strategies—original, oversampled, and undersampled. Each model was assessed using standard classification metrics, namely accuracy, recall, precision, and F1-score. Due to the critical nature

of sepsis detection, the recall metric received particular attention as it reflects the model's sensitivity to identifying true sepsis cases.

MODEL	Type	Recall_o	Recall_1	Precision_o	Precision_1	Accuracy
Logistic Regression	Original	1.00	0.06	0.93	0.68	0.93
	Oversampled	0.57	0.73	0.96	0.12	0.58
	Undersampled	0.57	0.76	0.97	0.12	0.58
Decision Tree	Original	0.96	0.52	0.96	0.49	0.93
	Oversampled	0.96	0.52	0.96	0.49	0.93
	Undersampled	0.96	0.52	0.96	0.49	0.93
Pruned Decision Tree	Original	0.99	0.48	0.96	0.72	0.95
	Oversampled	0.94	0.61	0.97	0.43	0.91
	Undersampled	0.82	0.78	0.98	0.26	0.82
Naive Bayes	Original	0.93	0.34	0.95	0.27	0.89
	Oversampled	0.88	0.52	0.96	0.25	0.85
	Undersampled	0.89	0.44	0.95	0.24	0.86
KNN	Original	1.00	0.10	0.93	0.85	0.93
	Oversampled	0.76	0.60	0.96	0.17	0.75
	Undersampled	0.88	0.36	0.95	0.19	0.84
Bagging	Original	0.99	0.50	0.96	0.78	0.95
	Oversampled	0.97	0.59	0.97	0.57	0.94
	Undersampled	0.87	0.82	0.98	0.33	0.87
Adaboost	Original	1.00	0.17	0.94	0.90	0.94
	Oversampled	0.98	0.49	0.96	0.65	0.94
	Undersampled	0.91	0.69	0.97	0.38	0.89
Gradient Boosting	Original	0.99	0.5	0.96	0.83	0.96
	Oversampled	0.97	0.61	0.97	0.63	0.95
	Undersampled	0.89	0.44	0.95	0.24	0.86
Random Forest	Original	0.99	0.40	0.96	0.85	0.95
	Oversampled	0.99	0.40	0.96	0.85	0.95
	Undersampled	0.99	0.40	0.96	0.85	0.95
Tuned RF	Original	1.00	0.30	0.95	0.87	0.95
	Oversampled	0.83	0.88	0.99	0.28	0.83
	Undersampled	0.83	0.88	0.99	0.28	0.83
Neural Network	Original	0.99	0.45	0.96	0.75	0.95
	Oversampled	0.74	0.57	0.96	0.14	0.72
	Undersampled	0.74	0.57	0.96	0.14	0.72
SVM	Original	0.93	0.58	0.97	0.38	0.90
	Oversampled	0.73	0.67	0.97	0.16	0.72
	Undersampled	0.93	0.58	0.97	0.38	0.90
XGBoost	Original	0.99	0.52	0.96	0.79	0.95
	Oversampled	0.99	0.57	0.97	0.76	0.96
	Undersampled	0.86	0.84	0.99	0.32	0.86

One common pattern we observe from this classification report table comparison is that we are getting good recall balances when class 1 precision is low, and we are getting bad recall balances when class 1 precision is relatively high.

The experimental results demonstrate that the *XGBoost* model trained on the *undersampled* dataset outperformed all other models in terms of recall balance. It achieved a recall of 0.86 for class 0 (non-sepsis) and 0.84 for class 1 (sepsis). However, the model achieved a precision of 0.99 for class 0 and 0.32 for class 1, accompanied by an overall accuracy of 0.86. While the precision for class 1 is relatively low than many other models, no other model has such a good recall balance like this one. Based on the case of sepsis early detection it is much more important to have a good recall balance.

Under under sampling, a pruned Decision Tree achieved recalls of 0.82 (non-sepsis) and 0.78 (sepsis), while Bagging reached 0.87 and 0.82. A tuned Random Forest on both resampled sets showed strong balance ($\text{Recall}_0 = 0.83$, $\text{Recall}_1 = 0.88$). Gradient Boosting on the oversampled data delivered excellent non-sepsis detection ($\text{Recall}_0 = 0.97$) and moderate sepsis sensitivity ($\text{Recall}_1 = 0.61$), and AdaBoost on under sampling yielded a more even split ($\text{Recall}_0 = 0.91$, $\text{Recall}_1 = 0.69$). In contrast, Logistic Regression and KNN struggled with sepsis detection without resampling ($\text{Recall}_1 \leq 0.10$), SVMs offered moderate balance (0.93/0.58), and neural nets-maintained $\text{Recall}_1 \approx 0.57$ but low precision, limiting clinical reliability.

So, we can say that while ensemble methods such as Random Forest and Bagging provided consistent performance across different class distributions, the *XGBoost* model trained on the undersampled dataset was identified as the most balanced and clinically applicable classifier. While we would love to achieve higher class 1 precision in the model, based on the models we tested Undersampled *XGBoost* delivered comparable sensitivity across both classes, coupled with high specificity, makes it the optimal candidate for deployment in intensive care environments where early and accurate sepsis prediction is essential.

4. Discussion

4.1 Real-World Implications

The undersampled *XGBoost* model's balanced recall (0.86 non-sepsis, 0.84 sepsis) suggests it could meaningfully reduce missed sepsis cases six hours before clinical onset. In practice, this early warning might enable timely interventions—such as antibiotic administration or hemodynamic support—potentially lowering ICU length of stay, morbidity, and mortality rates. High precision for non-sepsis (0.99) means few false alarms, limiting “alert fatigue” for frontline staff. However, modest sepsis precision (0.32) implies that roughly two-thirds of positive alerts could be false positives, risking unnecessary treatments or diagnostic workups and associated costs.

4.2 Limitations

While this study presents promising results, it is important to acknowledge several limitations that could influence the generalizability and reliability of the findings. First, the dataset utilized—although extensive—is derived from a specific patient population available through the PhysioNet Challenge dataset, which is based on select ICU centers. As such, model performance may vary when applied to different hospitals, healthcare systems, or geographical regions with diverse patient demographics (Johnson).

Second, the time-series nature of ICU data was not fully exploited in this study. Although the dataset included hourly measurements, the models used treated data as static snapshots. Consequently, potentially informative temporal patterns were not leveraged. Future research could benefit from employing sequence-

based models such as Long Short-Term Memory (LSTM) networks or Temporal Convolutional Networks (TCNs), which are known to capture temporal dependencies in clinical data (Shickel) (Choi).

Third, although class imbalance was addressed using SMOTE and random undersampling, each method introduces its own challenges. Oversampling may lead to overfitting due to synthetic data that lacks biological plausibility, while undersampling may omit useful information from the majority class. Recent studies recommend integrating cost-sensitive learning or hybrid sampling methods as alternatives for improving robustness (Fernández).

Interpretability remains a critical requirement for clinical adoption. While XGBoost offers built-in feature importance metrics, it lacks the transparency of white-box models such as logistic regression. Enhancing interpretability through methods like SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations) would provide clinicians with clearer insight into the factors influencing predictions, thereby increasing trust and usability (Lundberg).

Lastly, model validation was performed using internal cross-validation and a hold-out validation set. To ensure reliability in real-world settings, future work should include external validation across independent cohorts and prospective deployment studies within hospital workflows. Real-time system integration should also be evaluated for computational efficiency, latency, and user interface considerations.

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